

SkyMonitor V5 Capture-to-Export Architecture

Core Pipeline:

1. AllSkyCaptureService orchestrates capture loop.
2. CameraAdapterBase handles exposure, acquisition, processing.
3. ExposureAnalyzer feeds adjustments to controller.
4. RemoteFramePublisher forwards frames to remote clients.
5. FrameExportPublisher emits raw/processed envelopes.
6. BackgroundFrameStacker + FrameFilterPipeline process frames.
7. Export sinks (S3, filesystem) consume envelopes.
8. Telemetry store records metrics.

External Integrations:

- Camera SDK/Hardware (ZWO ASI, etc.)
- Remote consumers (WebSocket, HTTP, MQTT).
- Storage (S3, Filesystem).